

Chapter 14. Double Integrals over General Regions

Type I and Type II regions, reversing order, area, mass, and probability

Rectangles are the training ground for double integrals, but most meaningful regions are not rectangles. This chapter develops double integrals over curved and irregular plane regions. The main skill is translating a geometric region into correct limits of integration, choosing an efficient order, and interpreting the result as area, volume, mass, average value, or probability. A double integral over a general region has the same conceptual meaning as before:

$$\iint_D f(x, y) dA.$$

It adds the values of f over a two-dimensional region D . The new difficulty is that D may be bounded by curves, so the inner limits often depend on the outer variable.

Learning objectives

After studying this chapter, you should be able to:

1. describe Type I regions using inequalities $a \leq x \leq b$, $p(x) \leq y \leq q(x)$;
2. describe Type II regions using inequalities $c \leq y \leq d$, $g(y) \leq x \leq h(y)$;
3. set up and evaluate double integrals over nonrectangular regions;
4. sketch a region from integral limits and write integral limits from a sketch;
5. reverse the order of integration when it simplifies the computation;
6. compute area using $\iint_D 1 dA$;
7. split a complicated region into simpler pieces;
8. interpret double integrals as volume, mass, average value, and probability.

1 Why rectangles are not enough

In the previous chapter, the base region was usually a rectangle $R = [a, b] \times [c, d]$. For such a region, Fubini's theorem gives

$$\iint_R f(x, y) dA = \int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy.$$

But many natural regions are bounded by curves: disks, triangular feasible sets, regions under density curves, parameter constraints, geographic regions, and decision regions in classification.

For example, the region below the parabola $y = 4 - x^2$ and above the line $y = 3x$ is not a rectangle. Its vertical height changes as x changes.

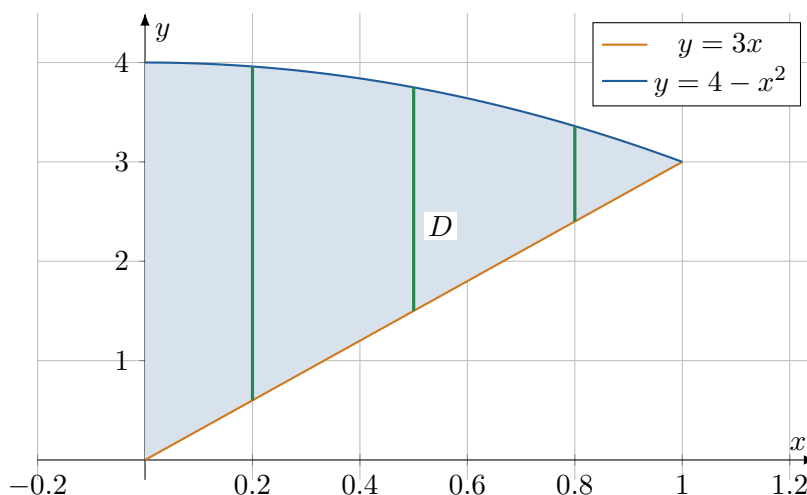


Figure 1: A nonrectangular region bounded by a parabola and a line. The vertical height of the region changes with x .

The key skill is to decide how to slice the region. A *vertical slice* leads to limits $p(x) \leq y \leq q(x)$. A *horizontal slice* leads to limits $g(y) \leq x \leq h(y)$. The same region may be easy in one direction and difficult in the other.

2 Type I regions: vertical slicing

A plane region D is called a *Type I region* if it can be described as

$$D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\},$$

where p and q are continuous functions and $p(x) \leq q(x)$ for $a \leq x \leq b$.

Geometrically, we move from left to right. For each fixed x , the variable y begins at the lower curve $y = p(x)$ and ends at the upper curve $y = q(x)$.

Double integral over a Type I region

If $D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\}$ and f is continuous on D , then

$$\iint_D f(x, y) dA = \int_a^b \int_{p(x)}^{q(x)} f(x, y) dy dx.$$

The inner integral adds along a vertical slice. The outer integral adds all slices from $x = a$ to $x = b$.

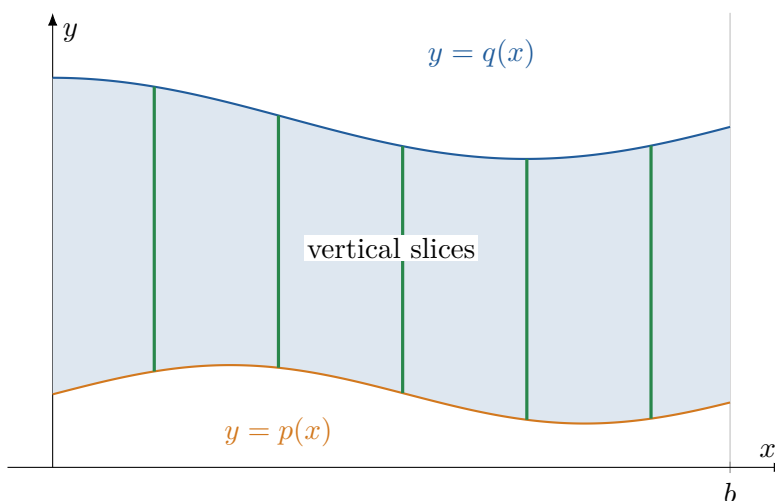


Figure 2: A Type I region is described by left-to-right limits and vertical slices.

Example 2.1: A region between a parabola and a line

Let D be the first-quadrant region bounded by $y = 4 - x^2$ and $y = 3x$. Compute

$$\iint_D 2xy \, dA.$$

Set the curves equal:

$$4 - x^2 = 3x \implies x^2 + 3x - 4 = 0 = (x + 4)(x - 1).$$

The first-quadrant intersection has $x = 1$ and $y = 3$. For $0 \leq x \leq 1$, the lower curve is $y = 3x$ and the upper curve is $y = 4 - x^2$. Thus

$$D = \{(x, y) : 0 \leq x \leq 1, 3x \leq y \leq 4 - x^2\}.$$

Therefore

$$\begin{aligned} \iint_D 2xy \, dA &= \int_0^1 \int_{3x}^{4-x^2} 2xy \, dy \, dx \\ &= \int_0^1 x[(4 - x^2)^2 - 9x^2] \, dx \\ &= \int_0^1 (16x - 17x^3 + x^5) \, dx = \frac{47}{12}. \end{aligned}$$

3 Type II regions: horizontal slicing

A plane region D is called a *Type II region* if it can be described as

$$D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\},$$

where g and h are continuous functions and $g(y) \leq h(y)$ for $c \leq y \leq d$.

Geometrically, we move from bottom to top. For each fixed y , the variable x begins at the left curve $x = g(y)$ and ends at the right curve $x = h(y)$.

Double integral over a Type II region

If $D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\}$ and f is continuous on D , then

$$\iint_D f(x, y) dA = \int_c^d \int_{g(y)}^{h(y)} f(x, y) dx dy.$$

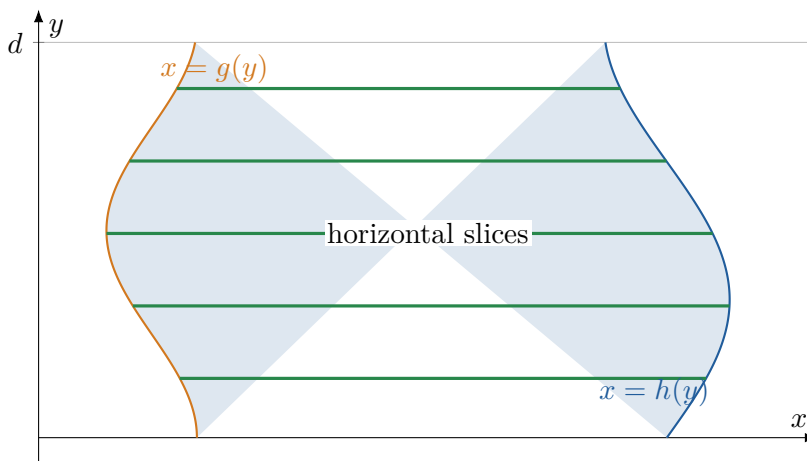


Figure 3: A Type II region is described by bottom-to-top limits and horizontal slices.

Example 3.1: A region between $x = y$ and $x = y^2$

Let D be the region bounded by $x = y$ and $x = y^2$. Compute

$$\iint_D (2xy + 6y^2) dA.$$

Intersections satisfy $y = y^2$, so $y = 0$ and $y = 1$. For $0 \leq y \leq 1$, we have $y^2 \leq x \leq y$. Hence

$$\begin{aligned} \iint_D (2xy + 6y^2) dA &= \int_0^1 \int_{y^2}^y (2xy + 6y^2) dx dy \\ &= \int_0^1 (7y^3 - 6y^4 - y^5) dy = \frac{23}{60}. \end{aligned}$$

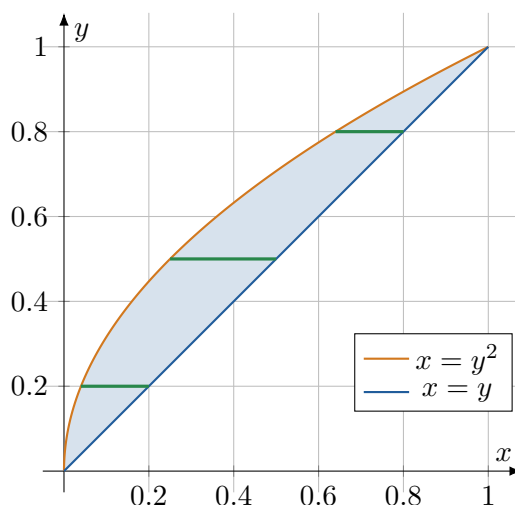


Figure 4: The region $y^2 \leq x \leq y$, $0 \leq y \leq 1$. Horizontal slices make the limits simple.

4 Choosing and reversing the order of integration

When setting up a double integral, first ask what a typical slice looks like. A vertical slice is useful if $x = \text{constant}$ enters and exits the region in one piece. A horizontal slice is useful if $y = \text{constant}$ enters and exits in one piece. Then ask which order makes the integrand easier.

Geometry first: algebra second

Many mistakes in double integrals are not integration mistakes. They are region mistakes. Before computing, check: the lower curve is really below the upper curve, the left curve is really left of the right curve, the intersection values are correct, and each slice stays inside the region.

Some iterated integrals are difficult in the given order but easy after reversing the order. Consider

$$\int_0^1 \int_y^1 e^{-x^2} dx dy.$$

The inner integral has no elementary antiderivative. The region is

$$D = \{(x, y) : 0 \leq y \leq 1, y \leq x \leq 1\}.$$

This is the triangle below $y = x$ inside the unit square. Rewriting in Type I form gives

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}.$$

Thus

$$\int_0^1 \int_y^1 e^{-x^2} dx dy = \int_0^1 \int_0^x e^{-x^2} dy dx = \int_0^1 x e^{-x^2} dx = \frac{1}{2}(1 - e^{-1}).$$

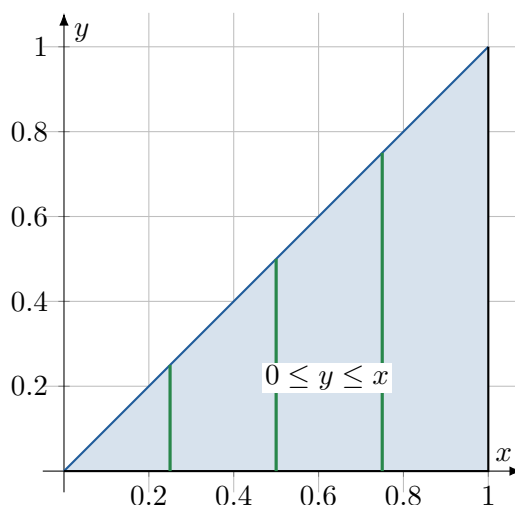


Figure 5: The triangle $0 \leq y \leq x \leq 1$. Reversing order turns a difficult integral into an elementary one.

Example 4.1: A reversed integral that requires splitting

Rewrite

$$\int_0^4 \int_{y/2}^{\sqrt{y+1}} f(x, y) \, dx \, dy$$

with the order $dy \, dx$. The given region is

$$0 \leq y \leq 4, \quad \frac{y}{2} \leq x \leq \sqrt{y+1}.$$

The boundaries become $y = 2x$ and $y = x^2 - 1$. A careful sketch shows that vertical slices change their lower or upper boundary, so the region splits:

$$\begin{aligned} \int_0^4 \int_{y/2}^{\sqrt{y+1}} f(x, y) \, dx \, dy &= \int_0^1 \int_0^{2x} f(x, y) \, dy \, dx \\ &\quad + \int_1^2 \int_{x^2-1}^{2x} f(x, y) \, dy \, dx \\ &\quad + \int_2^{\sqrt{5}} \int_{x^2-1}^4 f(x, y) \, dy \, dx. \end{aligned}$$

5 Area of a plane region

The area of a plane region D is computed by integrating the constant function 1:

Area by double integral

$$\text{Area}(D) = \iint_D 1 \, dA.$$

For a Type I region,

$$\text{Area}(D) = \int_a^b (q(x) - p(x)) dx.$$

For a Type II region,

$$\text{Area}(D) = \int_c^d (h(y) - g(y)) dy.$$

Example 5.1: Area between a line and a parabola

Find the area of the region bounded by $y = x - 1$ and $y^2 = 2x + 6$. Write both curves as x in terms of y :

$$y = x - 1 \iff x = y + 1, \quad x = \frac{y^2 - 6}{2}.$$

Intersections satisfy $y^2 = 2(y + 1) + 6$, so $(y - 4)(y + 2) = 0$. Thus $-2 \leq y \leq 4$. The parabola lies on the left and the line lies on the right:

$$\begin{aligned} \text{Area}(D) &= \int_{-2}^4 \int_{(y^2-6)/2}^{y+1} 1 \, dx \, dy \\ &= \int_{-2}^4 \left(y + 1 - \frac{y^2 - 6}{2} \right) dy = 18. \end{aligned}$$

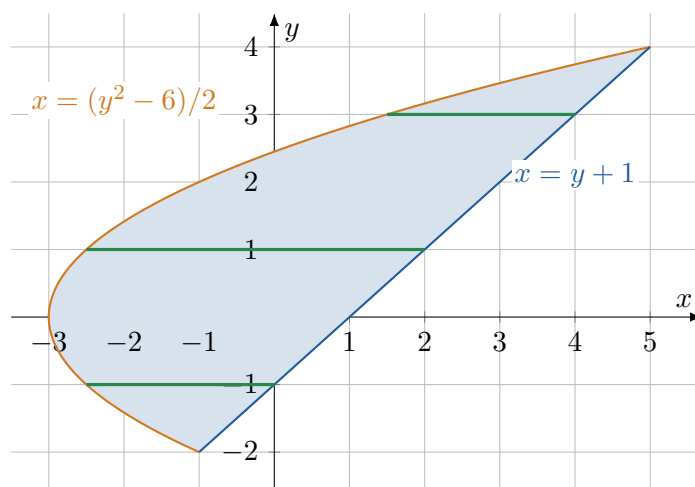


Figure 6: Area between the line $x = y + 1$ and the parabola $x = (y^2 - 6)/2$ by horizontal slices.

6 Splitting regions

Not every region is Type I or Type II in a single piece. Some regions must be split into simpler pieces. If $D = D_1 \cup D_2$ and the interiors do not overlap, then

$$\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA.$$

This is the two-dimensional version of splitting a one-variable integral at a point.

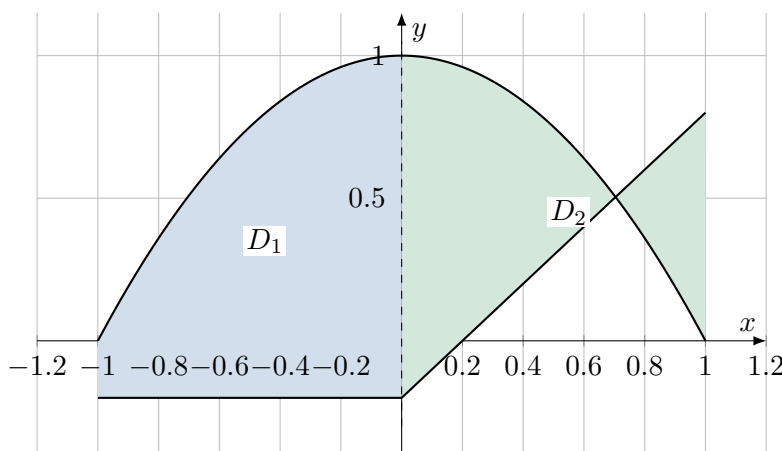


Figure 7: Some regions are best handled by splitting them into simpler Type I or Type II pieces.

When splitting is necessary

Splitting is needed when a single vertical or horizontal slice does not enter and exit the region through the same pair of boundary curves for the whole range. Splitting is not a failure; it is part of correct geometric modeling.

7 Volume under a surface over a curved base

If $f(x, y) \geq 0$ on a region D , then

$$V = \iint_D f(x, y) dA$$

is the volume of the solid above D and below the surface $z = f(x, y)$.

Example 7.1: Volume over a triangular region

Let D be the triangular region with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$. Find the volume under $z = 3 + x + y$ over D . The region is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 2x\}.$$

Therefore

$$\begin{aligned} V &= \int_0^1 \int_0^{2x} (3 + x + y) dy dx \\ &= \int_0^1 (6x + 4x^2) dx = 3 + \frac{4}{3} = \frac{13}{3}. \end{aligned}$$

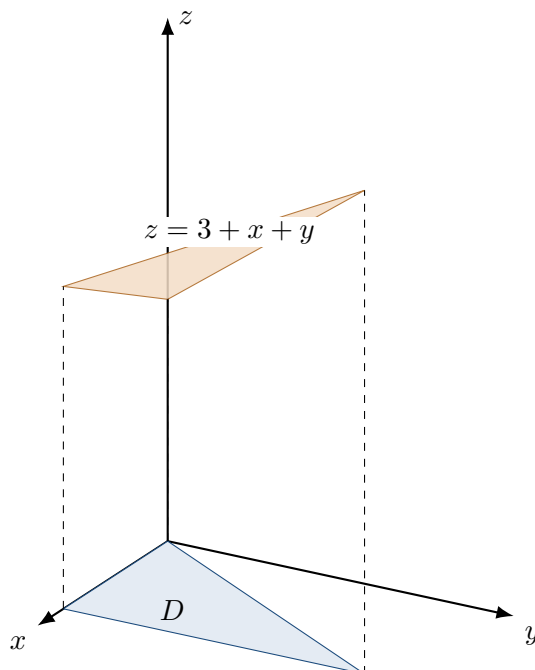


Figure 8: A surface over a triangular base region. The double integral gives volume.

8 Density, mass, and average value over general regions

If a thin plate occupies a region D and has density $\rho(x, y)$, then the mass is

$$m = \iint_D \rho(x, y) \, dA.$$

The average value of f over D is

$$f_{\text{avg}} = \frac{1}{\text{Area}(D)} \iint_D f(x, y) \, dA.$$

These formulas are the same as for rectangles, but now the geometry of D must be handled carefully.

Example 8.1: Average value on a triangle

Find the average value of $f(x, y) = x + 2y$ over

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}.$$

The area is $\text{Area}(D) = 1/2$. Compute

$$\begin{aligned} \iint_D (x + 2y) \, dA &= \int_0^1 \int_0^{1-x} (x + 2y) \, dy \, dx \\ &= \int_0^1 (1 - x) \, dx = \frac{1}{2}. \end{aligned}$$

Thus $f_{\text{avg}} = (1/2)/(1/2) = 1$.

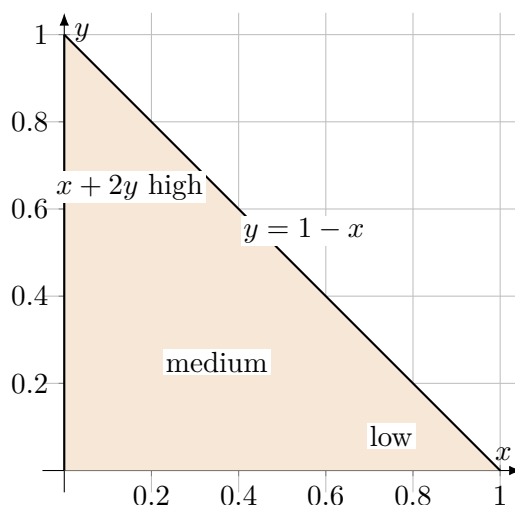


Figure 9: Average value over the triangular region $0 \leq x \leq 1$, $0 \leq y \leq 1 - x$.

9 Probability over a region

In statistics, a joint density $p(x, y)$ assigns probability density to points in the plane. If (X, Y) has joint density p , then

$$P((X, Y) \in D) = \iint_D p(x, y) dA.$$

This is one of the most important applications of double integrals over general regions.

Example 9.1: Probability over a curved event

Suppose (X, Y) is uniformly distributed over the unit square $[0, 1] \times [0, 1]$. Find $P(Y \leq X^2)$. The event is

$$D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}.$$

Since the density is 1 on the unit square,

$$P(Y \leq X^2) = \int_0^1 \int_0^{x^2} 1 dy dx = \int_0^1 x^2 dx = \frac{1}{3}.$$

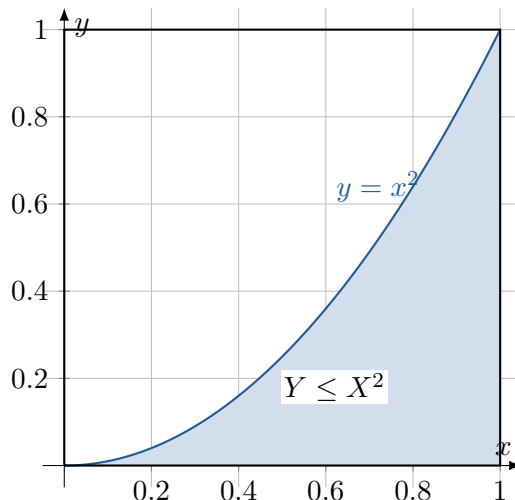


Figure 10: The probability $P(Y \leq X^2)$ for a uniform point in the unit square equals the area under $y = x^2$.

A nonuniform example: let $p(x, y) = 2$ on the triangle $T = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$ and 0 outside T . Since $\text{Area}(T) = 1/2$, this is a density. For the event $Y \leq X$,

$$P(Y \leq X) = \int_0^{1/2} \int_0^x 2 \, dy \, dx + \int_{1/2}^1 \int_0^{1-x} 2 \, dy \, dx = \frac{1}{2}.$$

10 Numerical integration over irregular regions

For complicated regions or complicated functions, exact integration may be difficult. Two common numerical strategies are grid-based approximation and Monte Carlo integration.

Suppose D lies inside a rectangle R of area $A(R)$. Then

$$\iint_D f(x, y) \, dA = \iint_R f(x, y) \mathbf{1}_D(x, y) \, dA,$$

where $\mathbf{1}_D(x, y)$ is 1 inside D and 0 outside D .

A Monte Carlo approximation samples random points in R :

$$\iint_D f(x, y) \, dA \approx A(R) \cdot \frac{1}{N} \sum_{i=1}^N f(X_i, Y_i) \mathbf{1}_D(X_i, Y_i).$$

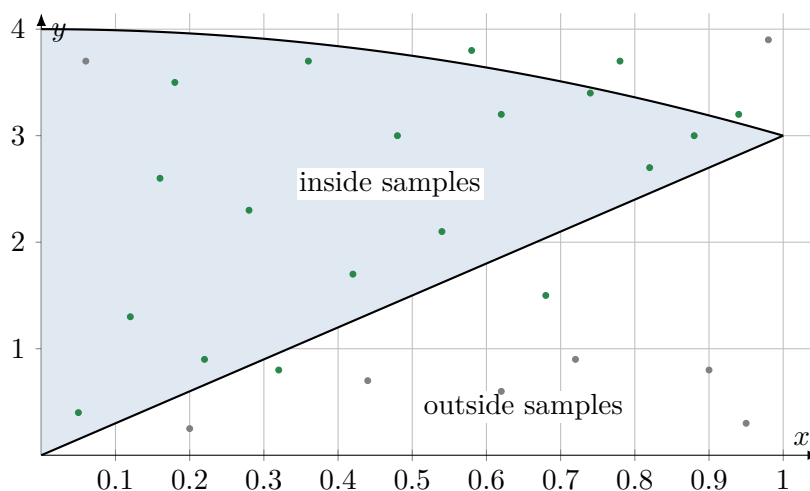


Figure 11: Monte Carlo integration over a curved region by sampling points in an enclosing rectangle.

Monte Carlo integration is not usually the most accurate method for low-dimensional smooth problems, but it scales naturally to high-dimensional problems. This is why it is central in statistics, Bayesian computation, and machine learning.

A common implementation mistake

When using numerical integration libraries, check the order of arguments. Some libraries call the integrand as $f(y, x)$ for an integral written in the order $dy dx$. The mathematical order and software argument order do not always look the same.

11 Checklist for setting up a double integral

1. Sketch the region and mark all boundary curves.
2. Find intersection points; these determine outer limits.
3. Choose vertical or horizontal slices. Decide whether the region is Type I, Type II, or must be split.
4. Write the inequalities. Do not compute until the region is correct.
5. Check lower/upper or left/right boundaries with a sample point.
6. Set up the integral, matching the inner variable to the inner differential.
7. Simplify before integrating; sometimes reversing order is better.
8. Interpret the result as area, volume, mass, average value, probability, or signed accumulation.

12 Modern applications

Statistics and probability

A joint density over a nonrectangular region appears when variables are constrained. Examples include $0 \leq y \leq x \leq 1$ for ordered random variables, $x + y \leq 1$ for compositional data, $y \leq x^2$ for nonlinear events, and feasible regions in Bayesian priors.

Machine learning

If $L(\theta_1, \theta_2)$ is a loss function and D is a feasible set, then

$$\frac{1}{\text{Area}(D)} \iint_D L(\theta_1, \theta_2) dA$$

is the average loss over D .

Applied mathematics

Regions may be determined by physical constraints: a membrane shape, a flow region, an obstacle boundary, a material interface, or a domain for a partial differential equation. Double integrals over such regions compute mass, energy, heat, charge, and probability.

13 Chapter summary

A double integral over a general region adds a function over a curved or constrained two-dimensional domain. The central task is translating geometry into limits.

- A Type I region has the form

$$D = \{(x, y) : a \leq x \leq b, p(x) \leq y \leq q(x)\},$$

and

$$\iint_D f(x, y) dA = \int_a^b \int_{p(x)}^{q(x)} f(x, y) dy dx.$$

- A Type II region has the form

$$D = \{(x, y) : c \leq y \leq d, g(y) \leq x \leq h(y)\},$$

and

$$\iint_D f(x, y) dA = \int_c^d \int_{g(y)}^{h(y)} f(x, y) dx dy.$$

- Area is computed by $\text{Area}(D) = \iint_D 1 dA$.
- Regions can be split: $\iint_D f dA = \iint_{D_1} f dA + \iint_{D_2} f dA$.
- Reversing order can turn a difficult integral into an elementary one.

The strongest students do not memorize limits. They learn to see slices.

14 Exercises

Conceptual exercises

1. In your own words, explain the difference between a Type I region and a Type II region.
2. Why do intersection points usually determine the outer limits of integration?
3. Explain why reversing the order of integration is a geometric operation before it is an algebraic operation.
4. Give an example of a region that is both Type I and Type II.
5. Give an example of a region that must be split to be described by vertical slices.
6. What does $\iint_D 1 \, dA$ compute?
7. If $f(x, y)$ is sometimes positive and sometimes negative on D , why is $\iint_D f \, dA$ not necessarily a volume?
8. Explain the role of the indicator function $\mathbf{1}_D(x, y)$ in numerical integration over an irregular region.

Skill exercises

9. Let $D = \{(x, y) : 0 \leq x \leq 2, x^2 \leq y \leq 2x\}$. Compute $\iint_D y \, dA$.
10. Let $D = \{(x, y) : 0 \leq y \leq 1, y \leq x \leq 2 - y\}$. Compute $\iint_D x \, dA$.
11. Find the area bounded by $y = x^2$ and $y = x + 2$.
12. Find the area bounded by $x = y^2$ and $x = 2 - y$.
13. Write the region bounded by $y = \sqrt{x}$ and $y = x^2$ as a Type I region and as a Type II region.
14. Evaluate $\int_0^1 \int_0^{1-x} (x + y) \, dy \, dx$.
15. Evaluate $\int_0^2 \int_{x/2}^1 xy \, dy \, dx$.
16. Set up, but do not evaluate, the volume under $z = x^2 + y^2$ over the region bounded by $y = x$ and $y = x^2$.

Reversing-order exercises

17. Reverse the order of integration: $\int_0^1 \int_0^x f(x, y) \, dy \, dx$.
18. Reverse the order of integration: $\int_0^4 \int_{\sqrt{y}}^2 f(x, y) \, dx \, dy$.
19. Evaluate by reversing order: $\int_0^1 \int_y^1 \cos(x^2) \, dx \, dy$.
20. Evaluate by reversing order: $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$.
21. Rewrite in the opposite order: $\int_0^1 \int_{x^2}^{\sqrt{x}} f(x, y) \, dy \, dx$.
22. Rewrite in the opposite order: $\int_0^1 \int_y^{2-y} f(x, y) \, dx \, dy$.

Applications and challenges

23. A plate occupies $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$ and has density $\rho(x, y) = 1 + x + y$. Find its mass.
24. Find the average value of $f(x, y) = xy$ over $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x\}$.
25. A point is chosen uniformly from the triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, 1)$. Find the probability that $Y \leq X$.
26. A point is chosen uniformly from the unit square. Find the probability that $X^2 \leq Y \leq X$.
27. Let D be the disk $x^2 + y^2 \leq 1$. Without using polar coordinates, explain why this region is not naturally Type I or Type II in one simple formula unless square roots are used.
28. Approximate $\iint_D e^{-(x^2+y^2)} dA$, where $D = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}$.
29. Let D be the region bounded by $y = x^2$ and $y = 2x - x^2$. Compute $\iint_D x dA$.
30. Let $D = \{(x, y) : 0 \leq x \leq 1, x^2 \leq y \leq x\}$. Compute the average value of $f(x, y) = x + y$ over D .
31. Find a region D and an integrand f for which one order of integration is elementary and the other is not.
32. Let D be the region bounded by $x = 0$, $y = 1$, and $y = x^2$. Set up $\iint_D f(x, y) dA$ in both orders.

15 Python lab: slicing, switching, and probability

In the companion Python lab, you will plot Type I and Type II regions, approximate double integrals by grid sums, reverse integration order from a plotted region, compute area and average value over triangular and parabolic regions, estimate probabilities by Monte Carlo simulation, and compare symbolic, numerical, and simulation-based answers.

16 Looking ahead

This chapter still uses rectangular coordinates. In the next chapter, we introduce polar coordinates and general changes of variables. That will make circular and radial regions much easier, and it will introduce one of the most important ideas in multivariable integration: the area element changes when coordinates change.